
Network Topology & VPN

1.Hyrje

Rrjetat kompjuterike janë një komponent thelbësor në funksionimin e organizatave moderne, duke mundësuar komunikimin dhe shkëmbimin e të dhënave ndërmjet pajisjeve dhe përdoruesve. Me rritjen e kërkesave për siguri dhe efikasitet, organizatat janë të detyruara të implementojnë struktura rrjetesh të segmentuara dhe të sigurta. Ky projekt paraqet dizenjimin dhe implementimin e një infrastrukture rrjeti për një kompani me disa departamente dhe komunikim të sigurt ndërmjet lokacioneve.

2.Objektivat

Objektivat kryesore të projektit janë:

- Dizenjimi i një rrjeti për një kompani me disa departamente
- Ndarja e rrjetit në VLAN për rritjen e sigurisë dhe performancës
- Implementimi i komunikimit ndërmjet VLAN-ve
- Konfigurimi i një lidhjeje ndërmjet zyrës qendrore dhe degës
- Aplikimi i politikave të sigurisë përmes ACL
- Implementimi i një VPN për komunikim të sigurt

3.Zhvillimi i punës

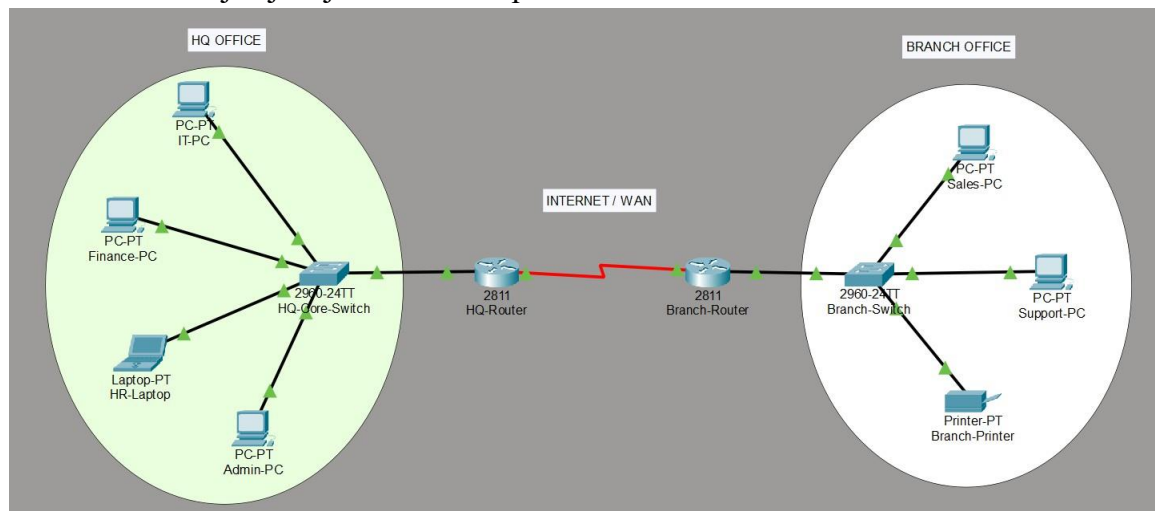
3.1 Përshkrimi i përgjithshëm i topologjisë

Në këtë projekt u ndërtua një rrjet kompjuterik që simulon një kompani reale me dy lokacione:

- HQ (Zyra Qendrore) - përmban disa departamente
- Branch (Degë) - përmban një rrjet të thjeshtë
- WAN (Internet) - lidh të dyja lokacionet

Topologjia është dizenuar në mënyrë që:

- të jetë e organizuar logjikisht
- të lehtësojë implementimin e sigurisë
- të simulojë një mjedis real enterprise



Network Topology & VPN

3.2 Konfigurimi bazë i pajisjeve

Fillimisht u realizua konfigurimi bazë i router-ave dhe switch-eve për të siguruar funksionimin korrekt të rrjetit.

Në router-at:

- u aktivizuan interface-t përkatëse duke përdorur komandën no shutdown
- u konfigurua lidhja WAN ndërmjet dy router-ave me adresim IP
- te njëri router u u konfiguruan hostname për identifikimin dhe administrimin e pajisjeve në rrjet

Në switch-et:

- u aktivizuan portat e përdorura
- u siguria që pajisjet fundore të jenë të lidhura korrekt

Pas konfigurimit, u realizua testimi i lidhjes WAN duke përdorur komandën ping, i cili rezultoi i suksesshëm.

The image shows two screenshots of Cisco IOS CLI interfaces. The top screenshot is for 'HQ-Router' and the bottom is for 'Branch-Router'. Both show the configuration of interfaces fa0/1 and s0/3/0, including enabling them and setting IP addresses. The HQ-Router configuration sets fa0/1 to 10.0.0.1 and s0/3/0 to 10.0.0.1. The Branch-Router configuration sets fa0/1 to 10.0.0.2 and s0/3/0 to 10.0.0.2. Both configurations include 'no shutdown' commands and show the state changes for the interfaces.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Fatbardh_Rama
Fatbardh_Rama(config)#interface fa0/1
Fatbardh_Rama(config-if)#no shutdown

Fatbardh_Rama(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
exit
Fatbardh_Rama(config)#interface s0/3/0
Fatbardh_Rama(config-if)#ip address 10.0.0.1 255.255.255.252
Fatbardh_Rama(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Fatbardh_Rama(config-if)#exit
Fatbardh_Rama(config)#end
Fatbardh_Rama#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
Fatbardh_Rama#
```

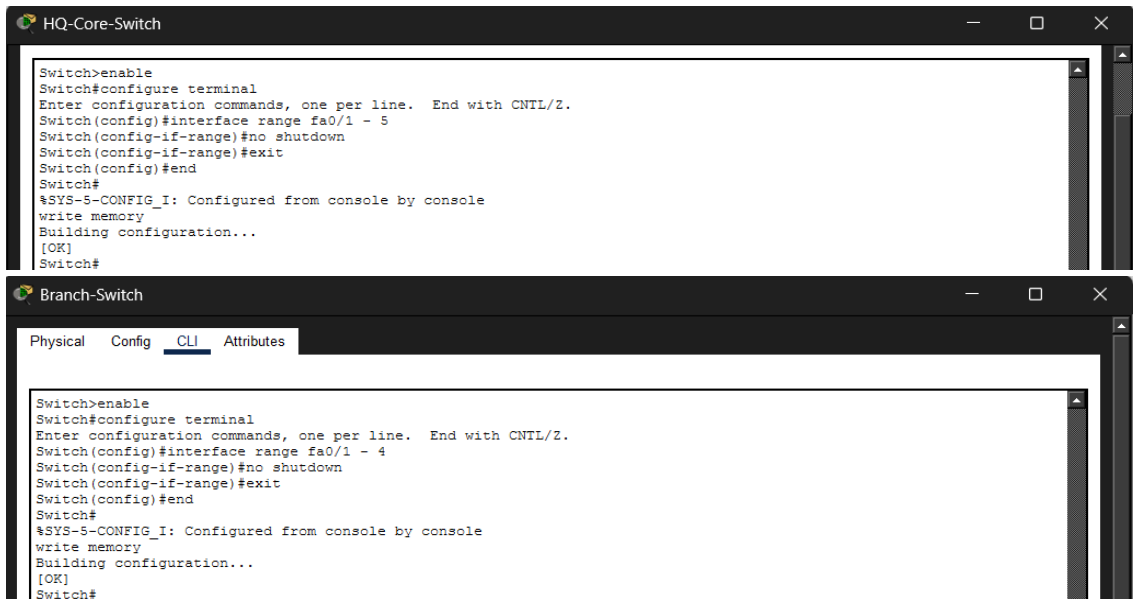
```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fa0/1
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
exit
Router(config)#interface s0/3/0
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
end
Router#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
Router#
```

Network Topology & VPN



- Switches

Testimi për të parë nëse funksionon:

```
Fatbardh_Rama>enable
Fatbardh_Rama#show ip interface brief
Interface                IP-Address      OK? Method Status          Protocol
FastEthernet0/0          unassigned      YES unset  administratively down down
FastEthernet0/1          unassigned      YES unset  up              up
Serial0/3/0              10.0.0.1        YES manual  up              up
Serial0/3/1              unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down
Fatbardh_Rama#

Fatbardh_Rama#ping 10.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/15/36 ms

Fatbardh_Rama#
```

Network Topology & VPN

3.3 Ndarja e rrjetit me VLAN

Në zyrën qendrore (HQ), rrjeti fizik u nda në disa rrjete logjike duke përdorur VLAN. Çdo departament u vendos në një VLAN të veçantë për të siguruar izolim të trafikut dhe rritje të sigurisë. VLAN (Virtual Local Area Network) përdoret për segmentimin logjik të rrjetit në grupe të ndara pajisjesh, edhe kur ato ndodhen në të njëjtën infrastrukturë fizike. Kjo metodë ndihmon në reduktimin e broadcast traffic, përmirësimin e performancës dhe rritjen e sigurisë së rrjetit duke izoluar departamentet nga njëri-tjetri. Për projektin u krijuan këto VLAN:

VLAN	Departamenti	Network
10	IT	192.168.10.0/24
20	Finance	192.168.20.0/24
30	HR	192.168.30.0/24
40	Admin	192.168.40.0/24

```
HQ-Core-Switch
Physical Config CLI Attributes
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name IT
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name FINANCE
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name HR
Switch(config-vlan)#exit
Switch(config)#vlan 40
Switch(config-vlan)#name ADMIN
Switch(config-vlan)#exit
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface fa0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 40
Switch(config-if)#exit
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
```

- Ndarja e rrjetit me VLAN & Trunking për *fa0/1*

Network Topology & VPN

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10	IT	active	Fa0/2
20	FINANCE	active	Fa0/3
30	HR	active	Fa0/4
40	ADMIN	active	Fa0/5
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#
```

- Kontrolli nëse ndarja u bë siç duhet me komandën *show vlan brief*

3.4 Inter-VLAN Routing (Router-on-a-Stick)

Për të mundësuar komunikimin ndërmjet VLAN-eve u përdor metoda router-on-a-stick.

Në këtë metodë:

- një interface fizik i routerit ndahet në disa subinterfaces
- çdo subinterface i përket një VLAN-i të caktuar
- secili ka një adresë IP që shërben si gateway për atë VLAN

Kjo bën të mundur që pajisjet në VLAN-e të ndryshme të komunikojnë ndërmjet tyre përmes routerit.

Te HQ-Router:

```
Fatbardh_Rama#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Fatbardh_Rama(config)#interface fa0/1
Fatbardh_Rama(config-if)#no shutdown
Fatbardh_Rama(config-if)#exit
Fatbardh_Rama(config)#interface fa0/1.10
Fatbardh_Rama(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.10, changed state to up

Fatbardh_Rama(config-subif)#encapsulation dot1Q 10
Fatbardh_Rama(config-subif)#ip address 192.168.10.1 255.255.255.0
Fatbardh_Rama(config-subif)#exit
Fatbardh_Rama(config)#interface fa0/1.20
Fatbardh_Rama(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.20, changed state to up

Fatbardh_Rama(config-subif)#encapsulation dot1Q 20
Fatbardh_Rama(config-subif)#ip address 192.168.20.1 255.255.255.0
Fatbardh_Rama(config-subif)#exit
Fatbardh_Rama(config)#interface fa0/1.30
Fatbardh_Rama(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.30, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.30, changed state to up

Fatbardh_Rama(config-subif)#encapsulation dot1Q 30
Fatbardh_Rama(config-subif)#ip address 192.168.30.1 255.255.255.0
Fatbardh_Rama(config-subif)#exit
Fatbardh_Rama(config)#interface fa0/1.40
Fatbardh_Rama(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.40, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.40, changed state to up

Fatbardh_Rama(config-subif)#encapsulation dot1Q 40
Fatbardh_Rama(config-subif)#ip address 192.168.40.1 255.255.255.0
Fatbardh_Rama(config-subif)#exit
Fatbardh_Rama(config)#
```

- Krijimi i subinterfaces

Network Topology & VPN

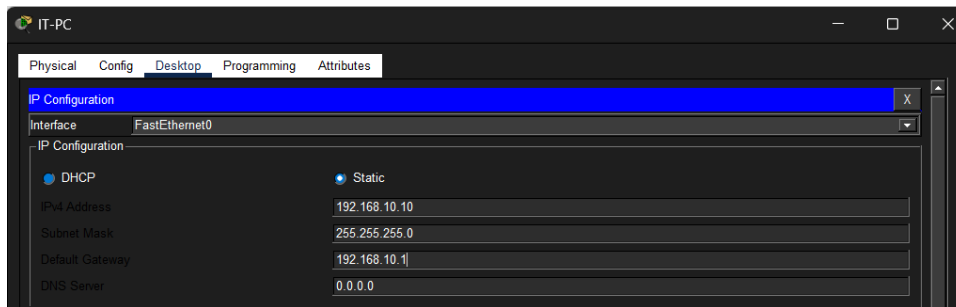
3.5 Testimi fillestar i komunikimit

Përpara konfigurimeve automatike, u vendosën adresat IP manualisht në pajisje për të testuar funksionimin e rrjetit.

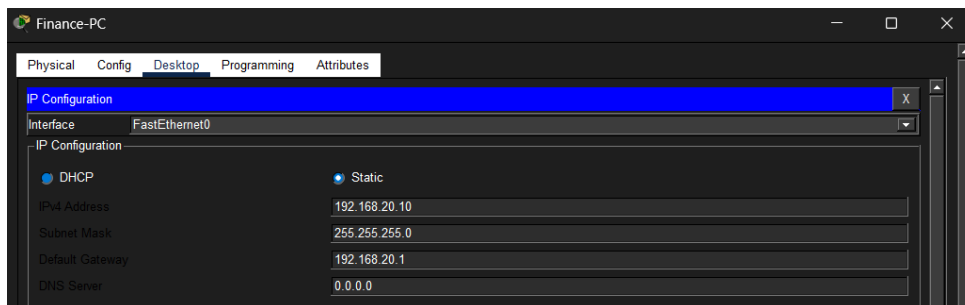
U realizuan teste komunikimi ndërmjet pajisjeve në VLAN të ndryshme dhe u konstatua se:

- komunikimi funksionon korrekt
- konfigurimi i VLAN dhe routing është i saktë

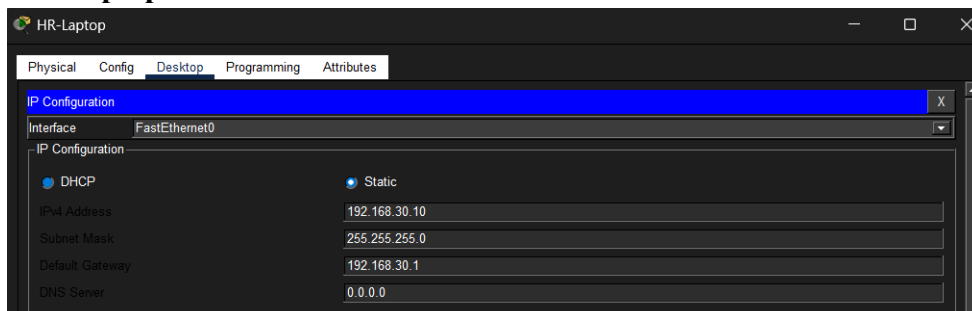
IT-PC



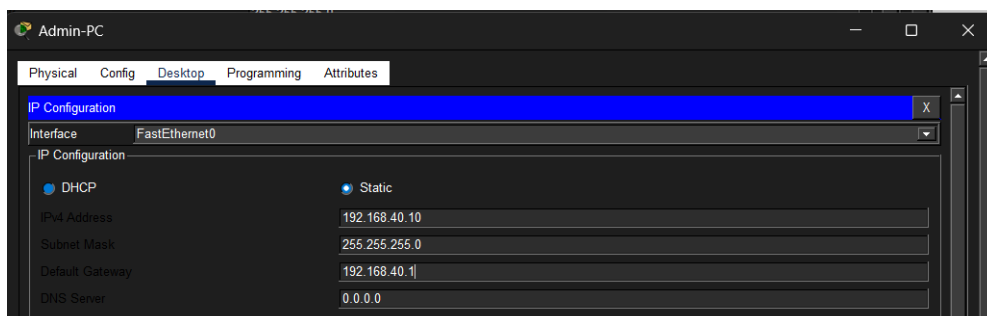
Finance-PC



HR-Laptop

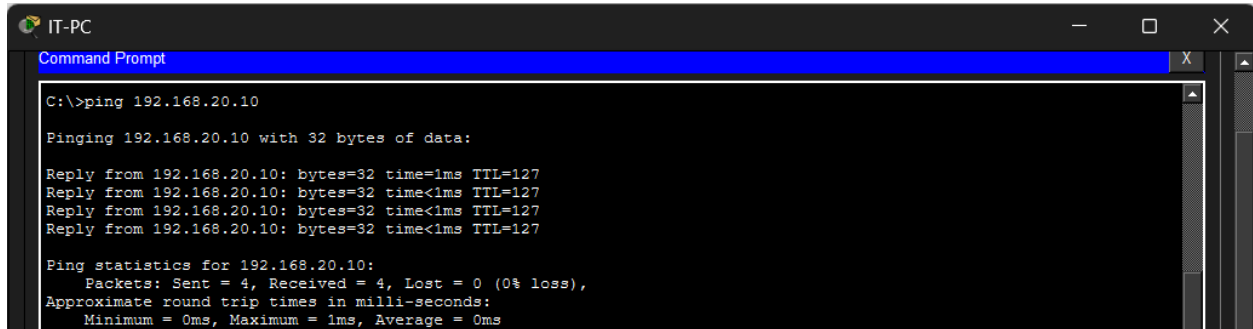


Admin-PC



Network Topology & VPN

Testimi:



```
IT-PC
Command Prompt

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

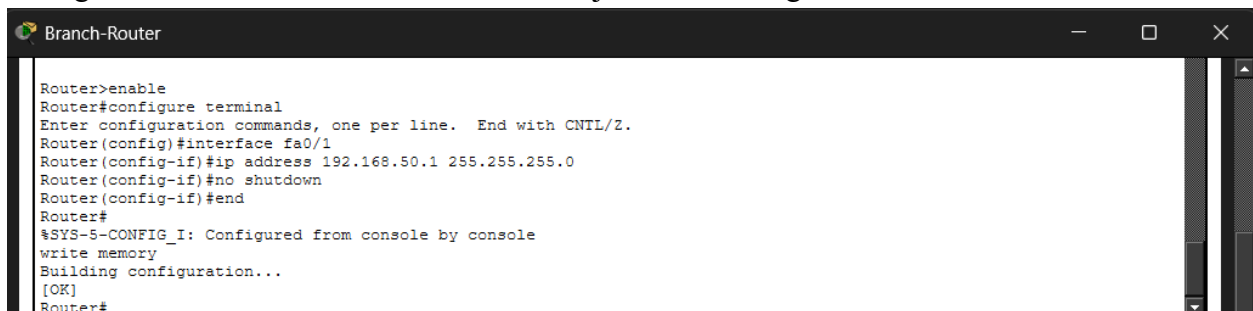
Reply from 192.168.20.10: bytes=32 time=1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

3.6 Konfigurimi i rrjetit në Branch

Në degë u implementua një rrjet i thjeshtë me adresim të veçantë.

Pajisjet u konfiguruan me IP manual dhe u vendos gateway përkatës që është routeri i degës. Ky konfigurim mundëson komunikimin brenda rrjetit lokal të degës.

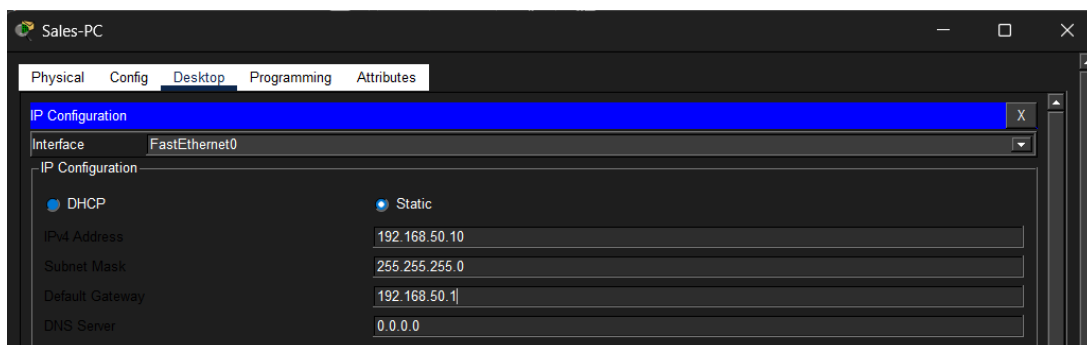


```
Branch-Router

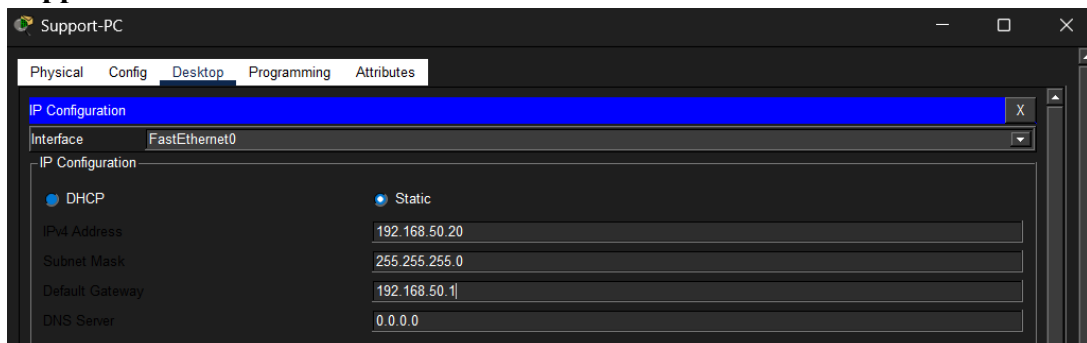
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fa0/1
Router(config-if)#ip address 192.168.50.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
Router#
```

Vendosja e IP manual në pajisjet e Branch:

Sales-PC



Support-PC



Network Topology & VPN

Branch-Printer

Port Status	☒ On
Link Speed	☒ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
MAC Address	000C.8532.5370
IP Configuration	
☐ DHCP	
☒ Static	
IPv4 Address	192.168.50.30
Subnet Mask	255.255.255.0

Global Settings	
Display Name Branch-Printer	
Gateway/DNS IPv4	
☐ DHCP	
☒ Static	
Default Gateway	192.168.50.1
DNS Server	

Testimi :

```
Sales-PC
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 192.168.50.20

Pinging 192.168.50.20 with 32 bytes of data:

Reply from 192.168.50.20: bytes=32 time<1ms TTL=128
Reply from 192.168.50.20: bytes=32 time<1ms TTL=128
Reply from 192.168.50.20: bytes=32 time<1ms TTL=128
Reply from 192.168.50.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.50.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.50.30

Pinging 192.168.50.30 with 32 bytes of data:

Reply from 192.168.50.30: bytes=32 time<1ms TTL=128
Reply from 192.168.50.30: bytes=32 time<1ms TTL=128
Reply from 192.168.50.30: bytes=32 time<1ms TTL=128
Reply from 192.168.50.30: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.50.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.50.1

Pinging 192.168.50.1 with 32 bytes of data:

Reply from 192.168.50.1: bytes=32 time<1ms TTL=255
Reply from 192.168.50.1: bytes=32 time<1ms TTL=255
Reply from 192.168.50.1: bytes=32 time<1ms TTL=255
Reply from 192.168.50.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.50.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Network Topology & VPN

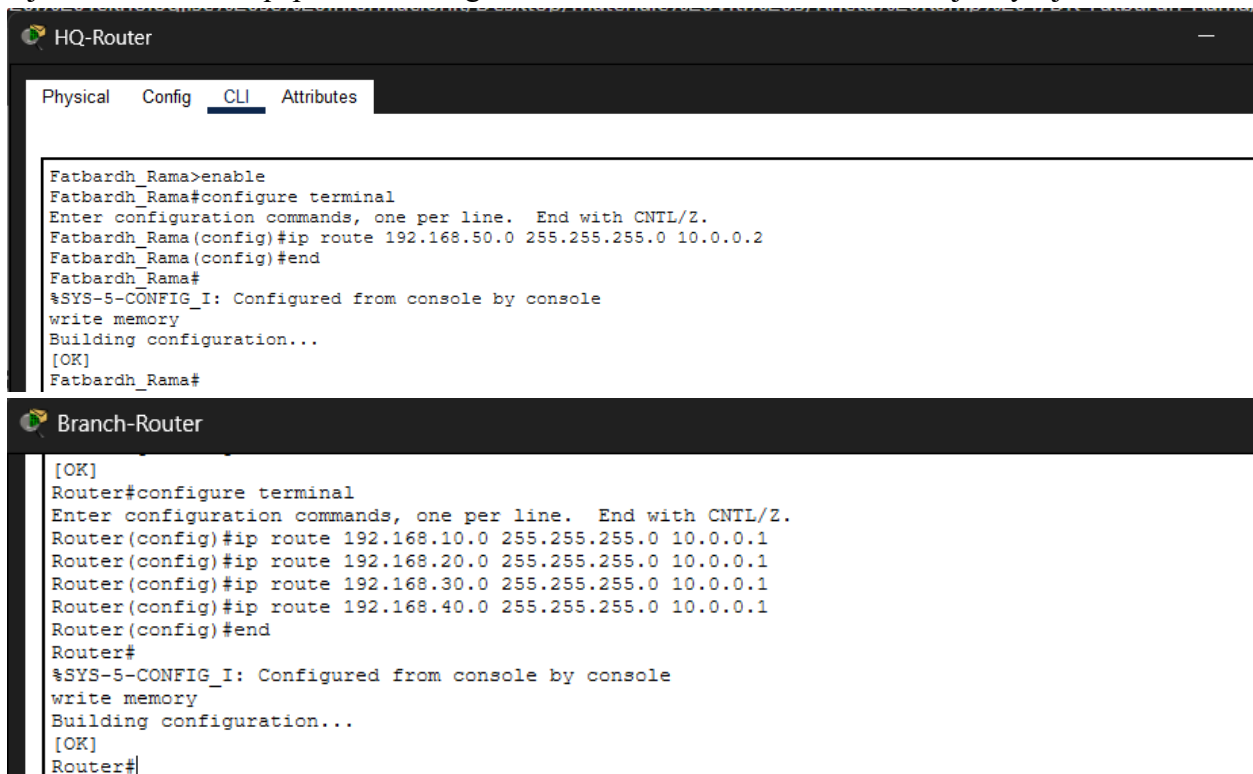
3.7 Routing ndërmjet HQ dhe Branch

Për të mundësuar komunikimin ndërmjet zyrës qendrore dhe degës u përdor routing statik.

Në këtë konfigurim:

- routeri i HQ njeh rrjetin e degës
- routeri i degës njeh VLAN-et e HQ

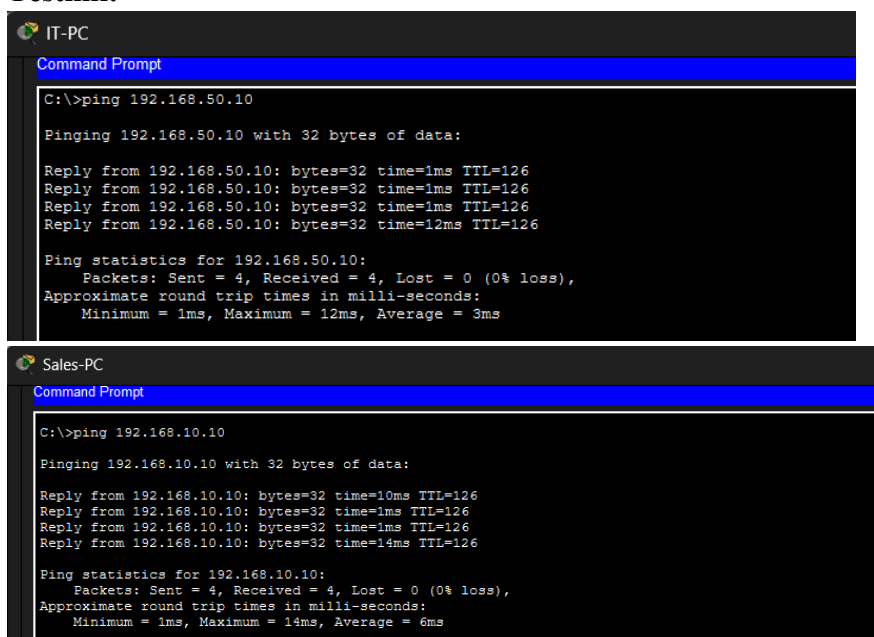
Kjo bën të mundur që paketat të dërgohen në destinacionin e duhur ndërmjet dy rrjeteve.



```
HQ-Router
Physical Config CLI Attributes
Fatbardh_Rama>enable
Fatbardh_Rama#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Fatbardh_Rama(config)#ip route 192.168.50.0 255.255.255.0 10.0.0.2
Fatbardh_Rama(config)#end
Fatbardh_Rama#
$SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
Fatbardh_Rama#

Branch-Router
[OK]
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 192.168.10.0 255.255.255.0 10.0.0.1
Router(config)#ip route 192.168.20.0 255.255.255.0 10.0.0.1
Router(config)#ip route 192.168.30.0 255.255.255.0 10.0.0.1
Router(config)#ip route 192.168.40.0 255.255.255.0 10.0.0.1
Router(config)#end
Router#
$SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
Router#
```

Testimi:



```
IT-PC
Command Prompt
C:\>ping 192.168.50.10

Pinging 192.168.50.10 with 32 bytes of data:

Reply from 192.168.50.10: bytes=32 time=1ms TTL=126
Reply from 192.168.50.10: bytes=32 time=1ms TTL=126
Reply from 192.168.50.10: bytes=32 time=1ms TTL=126
Reply from 192.168.50.10: bytes=32 time=12ms TTL=126

Ping statistics for 192.168.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 12ms, Average = 3ms

Sales-PC
Command Prompt
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time=10ms TTL=126
Reply from 192.168.10.10: bytes=32 time=1ms TTL=126
Reply from 192.168.10.10: bytes=32 time=1ms TTL=126
Reply from 192.168.10.10: bytes=32 time=14ms TTL=126

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 14ms, Average = 6ms
```

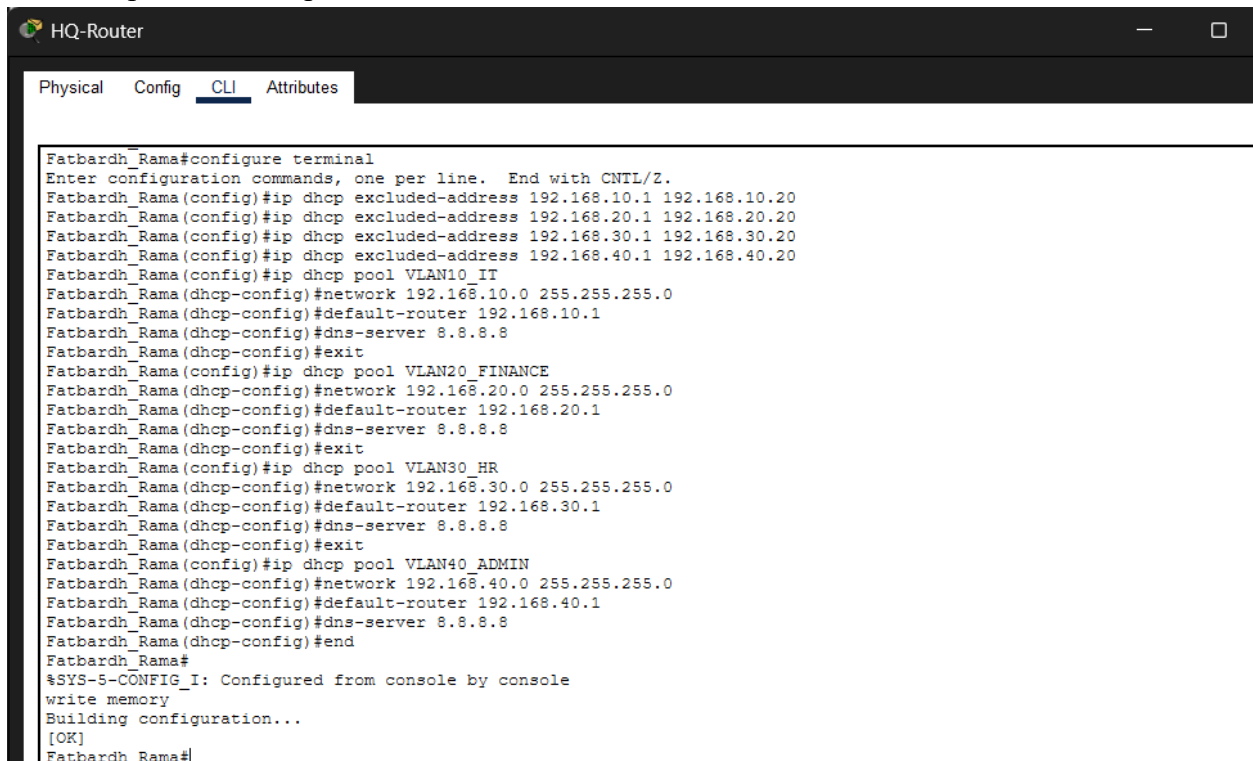
Network Topology & VPN

3.8 Implementimi i DHCP

DHCP (Dynamic Host Configuration Protocol) automatizon shpërndarjen e adresave IP dhe parametrave të rrjetit tek pajisjet klient. Përdorimi i DHCP redukton konfigurimet manuale dhe minimizon mundësinë e gabimeve në administrimin e rrjetit. Për të automatizuar konfigurimin e pajisjeve në rrjet u implementua DHCP në routerin kryesor. Për çdo VLAN u krijua një pool i veçantë, i cili përfshin:

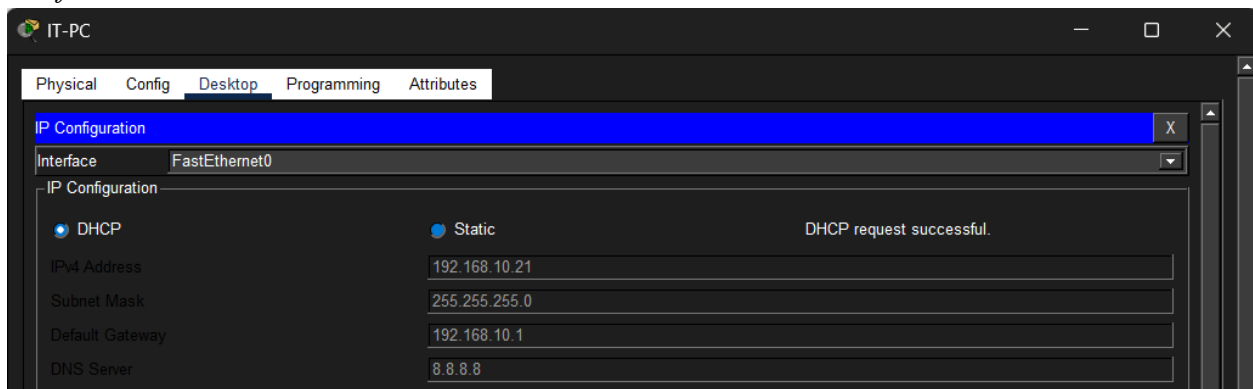
- adresat IP të disponueshme
- gateway
- DNS server

Pajisjet klient u konfiguruan të marrin IP automatikisht dhe rezultuan të marrin konfigurim korrekt sipas VLAN-it përkatës.

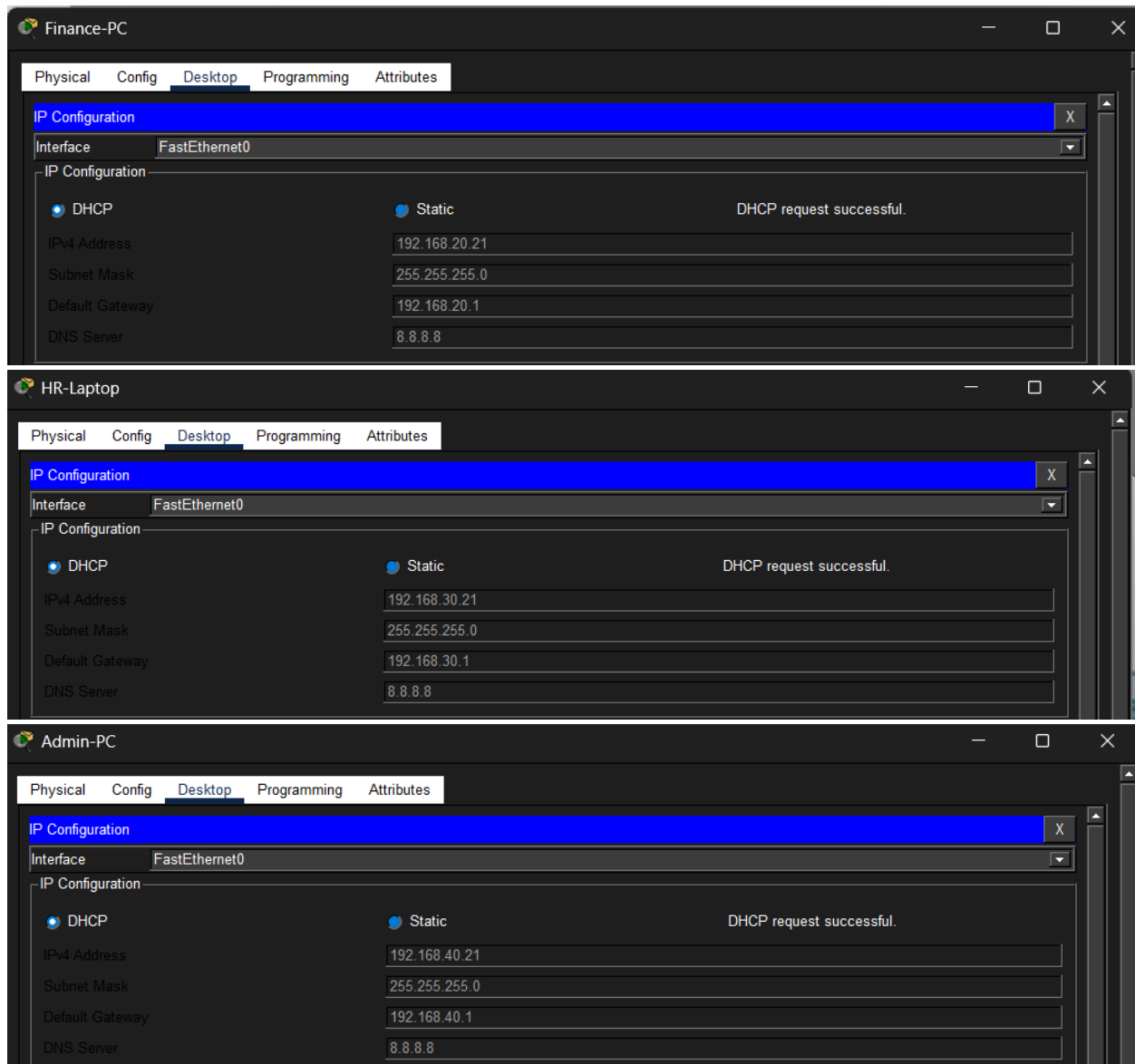
A screenshot of the HQ-Router CLI interface. The 'CLI' tab is selected. The terminal shows the configuration of four DHCP pools for VLANs 10, 20, 30, and 40. Each pool is named after its VLAN and has a specific network, default router, and DNS server configured. The configuration is saved to memory.

```
Fatbardh_Rama#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Fatbardh_Rama(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.20
Fatbardh_Rama(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.20
Fatbardh_Rama(config)#ip dhcp excluded-address 192.168.30.1 192.168.30.20
Fatbardh_Rama(config)#ip dhcp excluded-address 192.168.40.1 192.168.40.20
Fatbardh_Rama(config)#ip dhcp pool VLAN10_IT
Fatbardh_Rama(dhcp-config)#network 192.168.10.0 255.255.255.0
Fatbardh_Rama(dhcp-config)#default-router 192.168.10.1
Fatbardh_Rama(dhcp-config)#dns-server 8.8.8.8
Fatbardh_Rama(dhcp-config)#exit
Fatbardh_Rama(config)#ip dhcp pool VLAN20_FINANCE
Fatbardh_Rama(dhcp-config)#network 192.168.20.0 255.255.255.0
Fatbardh_Rama(dhcp-config)#default-router 192.168.20.1
Fatbardh_Rama(dhcp-config)#dns-server 8.8.8.8
Fatbardh_Rama(dhcp-config)#exit
Fatbardh_Rama(config)#ip dhcp pool VLAN30_HR
Fatbardh_Rama(dhcp-config)#network 192.168.30.0 255.255.255.0
Fatbardh_Rama(dhcp-config)#default-router 192.168.30.1
Fatbardh_Rama(dhcp-config)#dns-server 8.8.8.8
Fatbardh_Rama(dhcp-config)#exit
Fatbardh_Rama(config)#ip dhcp pool VLAN40_ADMIN
Fatbardh_Rama(dhcp-config)#network 192.168.40.0 255.255.255.0
Fatbardh_Rama(dhcp-config)#default-router 192.168.40.1
Fatbardh_Rama(dhcp-config)#dns-server 8.8.8.8
Fatbardh_Rama(dhcp-config)#end
Fatbardh_Rama#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
Fatbardh_Rama#
```

Verifikim:



Network Topology & VPN



3.9 Implementimi i ACL (Security Policies)

ACL (Access Control List) përdoret për kontrollin e trafikut në rrjet duke lejuar ose bllokuar komunikimin bazuar në adresa IP ose protokolle të caktuara. Implementimi i ACL-ve ndihmon në zbatimin e politikave të sigurisë dhe mbrojtjen e burimeve sensitive. Për të rritur sigurinë e rrjetit u implementuan politika kontrolli të aksesit (ACL). U vendosën rregulla që kufizojnë komunikimin ndërmjet departamenteve:

- departamenti Finance nuk ka akses në departamentin IT
- departamenti HR nuk ka akses në Finance

Këto rregulla sigurojnë që të dhënat sensitive të mos aksesohen nga departamente të tjera pa autorizim.

Testimet treguan se trafiku i ndaluar bllokohet, ndërsa trafiku i lejuar kalon normalisht.

Network Topology & VPN

```
HQ-Router
[OK]
Fatbardh_Rama#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Fatbardh_Rama(config)#access-list 100 deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
Fatbardh_Rama(config)#access-list 100 deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
Fatbardh_Rama(config)#access-list 100 permit ip any any
Fatbardh_Rama(config)#exit
Fatbardh_Rama#
%SYS-5-CONFIG_I: Configured from console by console

Fatbardh_Rama#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Fatbardh_Rama(config)#interface fa0/1.20
Fatbardh_Rama(config-subif)#ip access-group 100 in
Fatbardh_Rama(config-subif)#exit
Fatbardh_Rama(config)#interface fa0/1.30
Fatbardh_Rama(config-subif)#ip access-group 100 in
Fatbardh_Rama(config-subif)#exit
Fatbardh_Rama(config)#
```

- Krijimi dhe aplikimi i ACL

Testimi:

Finance-PC:

```
Finance-PC
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.21

Pinging 192.168.10.21 with 32 bytes of data:

Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.

Ping statistics for 192.168.10.21:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

HR-Laptop:

```
HR-Laptop
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.21

Pinging 192.168.20.21 with 32 bytes of data:

Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.

Ping statistics for 192.168.20.21:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

IT-PC:

```
C:\>ping 192.168.30.21

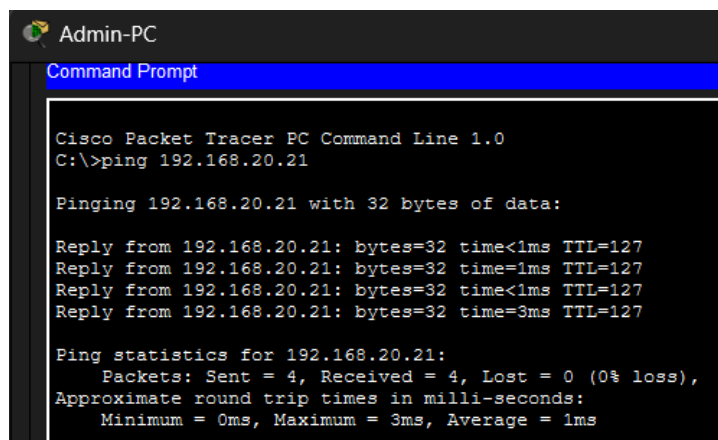
Pinging 192.168.30.21 with 32 bytes of data:

Reply from 192.168.30.21: bytes=32 time=11ms TTL=127
Reply from 192.168.30.21: bytes=32 time<1ms TTL=127
Reply from 192.168.30.21: bytes=32 time=11ms TTL=127
Reply from 192.168.30.21: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.30.21:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 5ms
```

Network Topology & VPN

Admin-PC:



```
Admin-PC
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.21

Pinging 192.168.20.21 with 32 bytes of data:

Reply from 192.168.20.21: bytes=32 time<1ms TTL=127
Reply from 192.168.20.21: bytes=32 time=1ms TTL=127
Reply from 192.168.20.21: bytes=32 time<1ms TTL=127
Reply from 192.168.20.21: bytes=32 time=3ms TTL=127

Ping statistics for 192.168.20.21:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 1ms
```

3.10 Implementimi i VPN

VPN (Virtual Private Network) krijon një tunel të sigurt komunikimi përmes një rrjeti publik. Në këtë projekt u implementua një VPN site-to-site për të siguruar enkriptimin e trafikut ndërmjet HQ dhe Branch, duke rritur konfidencialitetin dhe sigurinë e të dhënave.

Në fund u implementua një VPN site-to-site ndërmjet HQ dhe Branch për të siguruar komunikim të sigurt përmes rrjetit publik.

VPN u konfigurua duke përdorur:

- ISAKMP policy për autentikim dhe negociim
- pre-shared key për identifikim
- IPsec për enkriptimin e trafikut

Kjo krijon një tunel virtual të sigurt ndërmjet dy rrjeteve.

1. ISAKMP Policy

Te HQ-Router:

```
Fatbardh_Rama#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Fatbardh_Rama(config)#crypto isakmp policy 10
Fatbardh_Rama(config-isakmp)#encryption aes
Fatbardh_Rama(config-isakmp)#hash sha
Fatbardh_Rama(config-isakmp)#authentication pre-share
Fatbardh_Rama(config-isakmp)#group 2
Fatbardh_Rama(config-isakmp)#lifetime 86400
Fatbardh_Rama(config-isakmp)#
```

Te Branch-Router:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#hash sha
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 2
Router(config-isakmp)#lifetime 86400
Router(config-isakmp)#
```

Network Topology & VPN

2. Pre-shared key

Te HQ-Router:

```
Fatbardh_Rama(config-isakmp)#crypto isakmp key cisco123 address 10.0.0.2
Fatbardh_Rama(config)#
```

Te Branch-Router:

```
Router(config-isakmp)#crypto isakmp key cisco123 address 10.0.0.1
Router(config)#
```

3. ACL për trafikun që do enkriptohet

- Te HQ-Router

```
Fatbardh_Rama(config)#access-list 110 permit ip 192.168.0.0 0.0.255.255 192.168.50.0 0.0.0.255
Fatbardh_Rama(config)#
```

- Te Branch-Router

```
Router(config)#access-list 110 permit ip 192.168.50.0 0.0.0.255 192.168.0.0 0.0.255.255
Router(config)#
```

4. Transform set

```
Fatbardh_Rama(config)#crypto ipsec transform-set MYSET esp-aes esp-sha-hmac
Fatbardh_Rama(config)#
```

```
Router(config)#crypto ipsec transform-set MYSET esp-aes esp-sha-hmac
Router(config)#
```

- Kjo komandë u bë te të 2 routerat

5. Crypto map

```
Fatbardh_Rama(config)#crypto map VPNMAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Fatbardh_Rama(config-crypto-map)#set peer 10.0.0.2
Fatbardh_Rama(config-crypto-map)#set transform-set MYSET
Fatbardh_Rama(config-crypto-map)#match address 110
Fatbardh_Rama(config-crypto-map)#
```

- Te HQ-Router

```
Router(config)#crypto map VPNMAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Router(config-crypto-map)#set peer 10.0.0.1
Router(config-crypto-map)#set transform-set MYSET
Router(config-crypto-map)#match address 110
Router(config-crypto-map)#
```

- Te Branch-Router

6. Aplikimi në interface

```
Fatbardh_Rama(config-if)#crypto map VPNMAP
*Jan 3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Fatbardh_Rama(config-if)#
```

- Te HQ-Router

```
Router(config)#interface s0/3/0
Router(config-if)#crypto map VPNMAP
*Jan 3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router(config-if)#
```

- Te Branch-Router

Network Topology & VPN

Testimi :

```
Fatbardh_Rama#ping 192.168.50.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.50.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/11/22 ms

Fatbardh_Rama#
```

- Te HQ-Router

Verifikim:

```
Fatbardh_Rama#show crypto isakmp policy

Global IKE policy
Protection suite of priority 10
  encryption algorithm: AES - Advanced Encryption Standard (128 bit keys).
  hash algorithm:      Secure Hash Standard
  authentication method: Pre-Shared Key
  Diffie-Hellman group: #2 (1024 bit)
  lifetime:            86400 seconds, no volume limit

Fatbardh_Rama#
```

- Te HQ-Router

```
Router#show crypto map
Crypto Map VPNMAP 10 ipsec-isakmp
  Peer = 10.0.0.1
  Extended IP access list 110
    access-list 110 permit ip 192.168.50.0 0.0.0.255 192.168.0.0 0.0.255.255
  Current peer: 10.0.0.1
  Security association lifetime: 4608000 kilobytes/3600 seconds
  PFS (Y/N): N
  Transform sets={
    MYSET,
  }
  Interfaces using crypto map VPNMAP:
    Serial0/3/0

Router#
```

- Te Branch-Router

Teknologjitë dhe konceptet e përdorura në këtë projekt:

- Cisco Packet Tracer
- VLAN
- Inter-VLAN Routing
- Static Routing
- DHCP
- ACL
- VPN Site-to-Site
- TCP/IP
- WAN Connectivity

Network Topology & VPN

Konkluzioni

Në këtë projekt u realizua dizenjimi dhe implementimi i një rrjeti kompjuterik për një kompani me disa departamente dhe një degë të lidhur përmes një rrjeti WAN. Gjatë zhvillimit u përdorën koncepte kryesore të rrjeteve si VLAN, inter-VLAN routing, DHCP, ACL dhe VPN, duke krijuar një infrastrukturë të organizuar dhe funksionale.

Ndarja e rrjetit në VLAN mundësoi izolimin e trafikut ndërmjet departamenteve dhe rriti nivelin e sigurisë, ndërsa komunikimi ndërmjet tyre u realizua përmes metodës router-on-a-stick.

Implementimi i DHCP thjeshtoi konfigurimin e pajisjeve, duke lejuar shpërndarjen automatike të adresave IP.

Gjithashtu, përmes routing statik u siguria komunikimi ndërmjet zyrës qendrore dhe degës, ndërsa përdorimi i ACL kufizoi aksesin ndërmjet departamenteve sipas politikave të sigurisë. Në fund, implementimi i një VPN site-to-site siguroi një lidhje të mbrojtur dhe të enkriptuar ndërmjet dy rrjeteve.

Testimet e kryera treguan se rrjeti funksionon në mënyrë korrekte dhe përmbush të gjitha kërkesat e projektit. Ky implementim tregon rëndësinë e dizenjimit të mirë, segmentimit dhe sigurisë në ndërtimin e një rrjeti të besueshëm dhe efikas.